PAPER ID-311196

Subject Code: BCE503

Roll No: | | | | | | | | | | | | | |

BTECH

(SEM V) THEORY EXAMINATION 2025-26

QUANTITY ESTIMATION AND CONSTRUCTION MANAGEMENT

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 7 = 14**

Q No.	Question	CO	Level
a.	Differentiate between a preliminary estimate and a detailed estimate.	1	K2
b.	List the major components considered in the rate analysis of an item of work.	5	K2
c.	Explain any two types of contracts commonly used in construction projects.	2	K3
d.	What is the difference between PERT and CPM techniques?	2	K2
e.	What is cost slope, and how is it used in project time-cost optimization?	4	K2
f.	What is break-even analysis, and why is it important in project economics?	3	K3
g.	What is meant by owning cost and operating cost of construction equipment?	3	K2

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	Explain the long-wall and short-wall method of estimation with a suitable example.	1	K3
b.	Discuss the advantages and disadvantages of Lump-Sum Contracts and Item-Rate Contracts.	3	K3
c.	Write short notes on: (i) Owning and operating cost of equipment (ii) Earth-moving equipment	2	K3
d.	What is tender evaluation? Describe the main parameters considered during this process.	4	K2
e.	Describe the structure and applications of Bar Charts and Milestone Charts in project scheduling.	5	K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Describe in detail the centreline method of quantity estimation. Explain its advantages, limitations, and applications with a worked example.	1	K5
b.	What are the different types of estimates used in building construction? Explain each in detail along with their suitability.	1	K3

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Describe the complete procedure of rate analysis for a cement concrete work. Include material cost, labour, equipment, overheads, and contractor's profit.	2	K5
b.	Explain the various types of contracts used in construction projects. Compare any three in terms of risk distribution, payment mode, and applicability.		

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5. Attempt any *one* part of the following: 07 x 1 = 07

a.	Discuss the complete tendering process starting from tender notice to award of work. Explain pre-qualification, negotiation, and monitoring of contracts.	3	K4
b.	Explain PERT and CPM techniques. Compare them and discuss their application in project planning, scheduling, and resource allocation.	3	K3

6. Attempt any *one* part of the following: 07 x 1 = 07

a.	What is Work Breakdown Structure (WBS)? Describe in detail the steps in preparing a network from WBS and explain its use in construction management.	4	K4
b.	Explain equipment management in construction. Discuss productivity, owning cost, operating cost, and significance of simulation techniques in scheduling equipment.	4	K3

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	What are various project cost components? Explain budgeting, cost planning, total cost curve, cost slope, and break-even analysis in detail.	5	K3
b.	Discuss the methods of economic comparison—present worth method, equivalent annual cost method, and discounted cash flow method. Provide numerical illustrations for each.	5	K3

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